WORKED SOLUTION

STEP III 2002, Question 6

Pure • Nonlinear Differential Equations

STEP III 2002, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s3-q6

The key idea

The equation looks fearsome because of the fourth powers, but $y\frac{\mathrm{d}y}{\mathrm{d}x}$ is exactly half the derivative of y^2. So substitute $u = y^2 - 1$: the equation collapses to one about u and u' alone, which separates once we take a square root. The only care needed is with signs: the sign of u splits the plane into the strip $|y| < 1$ and the regions $|y| > 1$, and they give different families of curves.

Solution

Put $u = y^2 - 1$, so that $u' = 2y\frac{\mathrm{d}y}{\mathrm{d}x}$ and $y^4(\frac{\mathrm{d}y}{\mathrm{d}x})^4 = (\frac{1}{2}u')^4$. The equation becomes

$$\tfrac{1}{16}(u')^4 = u^2 \quad \Longleftrightarrow \quad (u')^2 = 4|u|,$$

taking the non-negative square root of both sides (both $(u')^2$ and $4|u|$ are non-negative, so nothing is lost).

Note first that a solution curve can never meet $y = 0$: there the left-hand side of the original equation is 0 but the right-hand side is 1. So on any solution through either given point, $y > 0$ throughout.

Ask yourself How do we solve $(u')^2 = 4u$ without worrying about which sign of u' to take?

Where $u > 0$ we may write $u = v^2$ with $v = \sqrt{u} > 0$. Then $u' = 2vv'$, so $(u')^2 = 4u$ becomes $4v^2(v')^2 = 4v^2$, i.e. $(v')^2 = 1$. Since v' is continuous and never zero it has a constant sign, so $v' = \pm 1$ and $v = \pm(x - a)$ for some constant a. Either way,

$$u = v^2 = (x - a)^2.$$

The sign ambiguity simply disappears on squaring. The same argument with $u = -v^2$ handles $u < 0$.

Part (i): through $(0, \frac{1}{2}\sqrt{3})$

Here $y^2 = \frac{3}{4} < 1$, so near this point $u < 0$, i.e. $|y| < 1$. Write $1 - y^2 = -u = v^2$ with $v > 0$; then $(u')^2 = -4u$ gives $(v')^2 = 1$ exactly as above, so $1 - y^2 = (x - a)^2$, that is

$$(x - a)^2 + y^2 = 1 :$$

circles of radius 1 centred on the x-axis. At $x = 0$ we need $a^2 = 1 - \frac{3}{4} = \frac{1}{4}$, so $a = \pm\frac{1}{2}$. Since $y > 0$, the solution curves through $(0, \frac{1}{2}\sqrt{3})$ are the upper semicircles

$$y = \sqrt{1 - (x - \tfrac{1}{2})^2} \qquad \text{and} \qquad y = \sqrt{1 - (x + \tfrac{1}{2})^2}.$$

Checking directly: on $(x - a)^2 + y^2 = 1$, implicit differentiation gives $y\frac{\mathrm{d}y}{\mathrm{d}x} = -(x - a)$, so $y^4(\frac{\mathrm{d}y}{\mathrm{d}x})^4 = (x - a)^4 = (1 - y^2)^2 = (y^2 - 1)^2$. ✓

Part (ii): through $(0, \frac{1}{2}\sqrt{5})$

Now $y^2 = \frac{5}{4} > 1$, so $u > 0$ and the argument above gives $y^2 - 1 = (x - a)^2$, that is

$$y^2 - (x - a)^2 = 1 :$$

rectangular hyperbolas with centre $(a, 0)$. At $x = 0$, $a^2 = \frac{5}{4} - 1 = \frac{1}{4}$, so again $a = \pm\frac{1}{2}$, and with $y > 0$ the solution curves are the upper branches

$$y = \sqrt{1 + (x - \tfrac{1}{2})^2} \qquad \text{and} \qquad y = \sqrt{1 + (x + \tfrac{1}{2})^2},$$

defined for all x. Directly: $y\frac{\mathrm{d}y}{\mathrm{d}x} = x - a$, so the left-hand side is $(x - a)^4 = (y^2 - 1)^2$. ✓

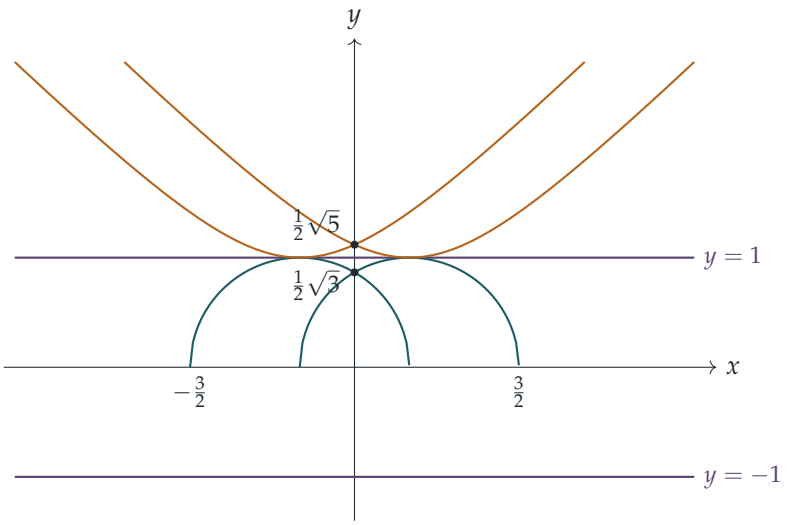
The lines $y = \pm 1$

If $y = 1$ or $y = -1$ for all x, then $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so the left-hand side is 0; and $y^2 - 1 = 0$, so the right-hand side is 0 too. Both lines are solutions.

Ask yourself How do these lines relate to the circles and hyperbolas?

Each circle $(x - a)^2 + y^2 = 1$ and each hyperbola $y^2 - (x - a)^2 = 1$ touches $y = 1$ at $(a, 1)$ with zero gradient (and likewise $y = -1$ at $(a, -1)$). So $y = \pm 1$ are the *envelopes* of both families, and they are not members of either family: they come from $u = 0$, which our division by v excluded. A consequence is that solutions are not unique through points on $y = 1$: for instance, following the circle with $a = \frac{1}{2}$ from $(0, \frac{1}{2}\sqrt{3})$ up to its top $(\frac{1}{2}, 1)$, then along $y = 1$, then up a hyperbola $y = \sqrt{1 + (x - b)^2}$ for $x \geq b$ (any $b \geq \frac{1}{2}$), gives a curve that is smooth enough (continuous gradient) and satisfies the equation everywhere. The four curves above are the "basic" solutions through the given points; every other solution through them is built from pieces of these, of $y = 1$, and of further circles and hyperbolas touching $y = 1$.

Sketch



The semicircles (teal) have centres $(\pm\frac{1}{2}, 0)$ and radius 1, meeting the x-axis vertically at $x = \pm\frac{1}{2} \pm 1$ and touching $y = 1$ at $x = \pm\frac{1}{2}$. The hyperbola branches (amber) touch $y = 1$ at the same two points and have asymptotes $y = \pm(x \mp \frac{1}{2})$, i.e. they grow like $|x|$. Each pair is symmetric about the y-axis, and the two curves of each pair cross on it at the given point.

Checks

- At $(0, \frac{1}{2}\sqrt{3})$ on the circle with $a = \frac{1}{2}$: $y' = -(x - a)/y = 1/\sqrt{3}$, so the left-hand side is $\frac{9}{16} \cdot \frac{1}{9} = \frac{1}{16}$ and the right-hand side is $(\frac{3}{4} - 1)^2 = \frac{1}{16}$.
- Numerically integrating $y' = \pm\sqrt{|y^2 - 1|}/y$ from $(0, \frac{1}{2}\sqrt{5})$ with the plus sign gives $y(1.5) = 2.2360680$, matching $\sqrt{1 + 2^2} = \sqrt{5}$ on the hyperbola with $a = -\frac{1}{2}$.

Common slips

- Taking the square root as $(u')^2 = 4u$ in both regions; for $|y| < 1$ this has no real solutions, and the circles are lost.
- Keeping only one sign of the square root at each stage, and so finding only one curve through each point instead of two.
- Forgetting that $y = \pm 1$ are not in either family, even though they touch every member.

Beyond STEP

The lines $y = \pm 1$ are *singular solutions*: envelopes of the general solution that no choice of the constant a produces. They appear exactly where the uniqueness theorem for first-order equations (Picard–Lindelöf) fails, because $\sqrt{|y^2 - 1|}$ is not Lipschitz at $y = \pm 1$, and that is why solutions can branch there. Envelopes of this kind are typical of first-order equations that are not linear in y' (Clairaut's equation is the classic example); Boyce and DiPrima, *Elementary Differential Equations*, discusses existence and uniqueness and when it breaks down.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.